

COMSATS University Islamabad

Lahore Campus – Department of Computer Science

Lab Assignment 03 – Spring 2026

Student Performance Analysis

Student Name: **Abu Huraira**

Roll Number: **Sp24-BAI-002**

Course: **AIC270 - Programming for Artificial Intelligence**

Instructor: **Mr. Shahrukh Naeem**

Semester: **5th**

Date: **08-05-2026**

Total Marks: **100**

OBJECTIVE: Learn professional Git workflow and reproducibility.

Steps Completed:

1. Created project folder: Student_Performance_Analysis (Lab_Assignment3)
2. Initialized Git repository using: git init
3. Created .gitignore file (ignores: archive.zip, __pycache__, .ipynb_checkpoints)
4. Created README.md with student details and project description
5. Created Jupyter notebook, loaded dataset, showed basic information
6. Made 4+ meaningful commits:
 - e1c0340 Initial commit: project setup with dataset and README
 - 781743a Add completed analysis notebook with all task outputs
 - 564716b branch-a: update course version to 1.0
 - bf3d799 branch-b: update course version to 2.0
 - f969fbc Resolve merge conflict between branch-a and branch-b
7. Created branch feature/statistics, added describe() statistics, merged into main
8. Simulated merge conflict using branch-a and branch-b on README.md
 - branch-a changed course line to: Version 1.0
 - branch-b changed same line to: Version 2.0
 - Conflict markers appeared in README.md (<<<<<< HEAD ... >>>>>>)
 - Resolved by keeping Version 2.0 and committing the fix
9. Pushed all branches to public GitHub repository:
https://github.com/abuhurairashabbir1/Student_Performance_Analysis

✓ Student Performance Analysis

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✓ Common Setup – Import Libraries & Load Dataset

[3]

✓ 0s

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

 Student_Performance_Analysis Public

 Pin

 Watch 0

 Fork 0

 Star 0

 main

 1 Branch

 0 Tags

Q Go to file

T

Add file

<> Code

 abuhurairashabbir1	Initial commit: project setup with dataset and README	e1c0340 · 16 minutes ago	🕒 1 Commit
data	Initial commit: project setup with dataset and README	16 minutes ago	
.gitignore	Initial commit: project setup with dataset and README	16 minutes ago	
README.md	Initial commit: project setup with dataset and README	16 minutes ago	

📖 README

Student Performance Analysis

About

No description, website, or topics provided.

📖 Readme

📈 Activity

☆ 0 stars

👁 0 watching

🍴 0 forks

Releases

No releases published

[Create a new release](#)

```
PS D:\uni\pf-ai\Lab_Assignment3> git commit -m "Add completed analysis notebook with all task outputs"
[main 781743a] Add completed analysis notebook with all task outputs
10 files changed, 2579 insertions(+)
create mode 100644 output/Screenshot 2026-05-08 155440.png
create mode 100644 output/best_visualization.png
create mode 100644 output/cleaned_students.csv
create mode 100644 output/correlation_heatmap.png
create mode 100644 output/messy_students.csv
create mode 100644 output/plot1_histogram.png
create mode 100644 output/plot2_boxplot.png
create mode 100644 output/plot3_scatter.png
create mode 100644 output/plot4_barchart.png
create mode 100644 student_performance_analysis.ipynb
PS D:\uni\pf-ai\Lab_Assignment3>
```

```
PS D:\uni\pf-ai\Lab_Assignment3> git log --oneline --graph --all
*   f969fbc (HEAD -> main, origin/main) Resolve merge conflict between branch-a and branch-b
| \
| * bf3d799 (branch-b) branch-b: update course version to 2.0
* | 564716b (branch-a) branch-a: update course version to 1.0
| /
* 781743a Add completed analysis notebook with all task outputs
* e1c0340 Initial commit: project setup with dataset and README
PS D:\uni\pf-ai\Lab_Assignment3> |
```

Mean | Median | Mode | Std | Min | Max

	Mean	Median	Mode	Std	Min	Max
Math Score	66.09	66.0	65.0	15.16	0.0	100.0
Reading Score	69.17	70.0	72.0	14.6	17.0	100.0
Writing Score	68.05	69.0	74.0	15.2	10.0	100.0

MEAN vs MEDIAN COMPARISON:

MATH SCORE:

Mean = 66.09

Median = 66.00

Diff = 0.09 → slightly right-skewed (mean > median)

READING SCORE:

Mean = 69.17

Median = 70.00

Diff = -0.83 → slightly left-skewed (mean < median)

WRITING SCORE:

Mean = 68.05

Median = 69.00

Diff = -0.95 → slightly left-skewed (mean < median)

CONCLUSION:

The mean and median values are very close for all three subjects, which means the score distributions are approximately symmetric. There is no heavy skew, indicating a balanced dataset. Math scores show the largest gap between mean and median, suggesting slightly more variation in math performance.

	math score	reading score	writing score
female	63.63	72.61	72.47
male	68.73	65.47	63.31

	math score	reading score	writing score
associate's degree	67.88	70.93	69.9
bachelor's degree	69.39	73.0	73.38
high school	62.14	64.7	62.45
master's degree	69.75	75.37	75.68
some college	67.13	69.46	68.84
some high school	63.5	66.94	64.89

OBSERVATION 1 – GENDER DIFFERENCES IN SUBJECTS:

Female students score higher in reading (avg ~72) and writing (avg ~72), while male students score higher in math (avg ~69).
This shows a clear subject-based performance difference between genders.

OBSERVATION 2 – PARENTAL EDUCATION MATTERS:

Students whose parents hold a master's degree score the highest on average across all subjects, while students with parents who only completed some high school score the lowest. Education level at home influences student academic performance significantly.

OBSERVATION 3 – BALANCED SCORE DISTRIBUTION:

The mean and median are very close for all three subjects, meaning the score distributions are symmetric with no extreme outliers.
This tells us the dataset is fairly representative and balanced.

OBSERVATION 4 – READING AND WRITING ARE CLOSELY LINKED:

Students who perform well in reading almost always perform well in writing.
This strong relationship makes sense as both skills are language-based.

OBSERVATION 5 – MATH HAS THE MOST VARIATION:

Math scores have the highest standard deviation among the three subjects, meaning student performance in math is more spread out (some very high, some very low), compared to reading and writing which are more consistent.

OBJECTIVE: Learn to clean messy real-world data.

STEP 1 – MESSY DATASET CREATED (messy_students.csv):

Problems intentionally added to the original dataset:

- Missing values added randomly to score columns (5% of rows)
- 20 duplicate rows added
- 3 outliers added: math=200, reading=-10, writing=999
- Messy dataset shape: 1020 rows x 8 columns

STEP 2 – PROBLEMS FOUND:

- Missing values: math=70 | reading=70 | writing=70
- Duplicate rows: 28
- Outliers (score < 0 or > 100):
 - math score: 1 outlier(s)
 - reading score: 1 outlier(s)
 - writing score: 1 outlier(s)

STEP 3 – CLEANING PROCESS:

1. Outliers fixed → replaced values outside 0-100 with NaN
2. Missing values filled → replaced NaN with median of each column
3. Duplicates removed → used drop_duplicates()
4. Index reset after cleaning

STEP 4 – BEFORE vs AFTER SUMMARY:

Rows:	Before = 1020		After = 992
Missing:	Before = 210		After = 0
Duplicates:	Before = 28		After = 0
Outliers:	Before = 3		After = 0

Cleaned dataset saved to: data/cleaned_students.csv

[16]

✓ 0s

```
# Step 2: Load messy dataset and show problems
messy_df = pd.read_csv('messy_students.csv')

print('=== Messy Dataset Info ===')
messy_df.info()
```

▼

```
... === Messy Dataset Info ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1020 entries, 0 to 1019
Data columns (total 8 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   gender                               1020 non-null   object
 1   race/ethnicity                       1020 non-null   object
 2   parental level of education          1020 non-null   object
 3   lunch                                1020 non-null   object
 4   test preparation course              1020 non-null   object
 5   math score                           950 non-null    float64
 6   reading score                        950 non-null    float64
 7   writing score                         950 non-null    float64
dtypes: float64(3), object(5)
memory usage: 63.9+ KB
```

[17]

✓ Os



```
print( \n--- outliers (scores outside 0-100) --- )  
for col in score_cols:  
    outliers = messy_df[(messy_df[col] < 0) | (messy_df[col] > 100)]  
    print(f'{col}: {len(outliers)} outlier(s)')
```

▼

... === Missing Values ===

gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	70
reading score	70
writing score	70
dtype: int64	

=== Duplicate Rows ===

Number of duplicates: 28

=== Outliers (scores outside 0-100) ===

math score: 1 outlier(s)
reading score: 1 outlier(s)
writing score: 1 outlier(s)

18]

✓ 0s



```
# Fill missing values with median
for col in score_cols:
    median_val = cleaned_df[col].median()
    cleaned_df[col].fillna(median_val, inplace=True)
print('Missing values filled with median.')

# Remove duplicates
cleaned_df.drop_duplicates(inplace=True)
cleaned_df.reset_index(drop=True, inplace=True)
print('Duplicates removed.')
```

✓

... Outliers fixed.

Missing values filled with median.

Duplicates removed.

/tmp/ipykernel_2855/2868152207.py:12: FutureWarning: A value is trying to be set on a copy of a Dat
The behavior will change in pandas 3.0. This inplace method will never work because the intermediat

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, i

```
cleaned_df[col].fillna(median_val, inplace=True)
```

	math score	reading score	writing score
group A	62.26	65.12	63.12
group B	63.76	67.63	65.83
group C	64.81	69.34	68.2
group D	67.13	70.03	70.15
group E	73.86	73.27	71.62

	math score	reading score	writing score
completed	69.77	73.89	74.42
none	64.29	66.85	64.86

	math score	reading score	writing score
math score	1.0	0.81	0.8
reading score	0.81	1.0	0.95
writing score	0.8	0.95	1.0

```
[25]  # GroupBy: Test Preparation Course
✓ 0s print('=== Average Scores by Test Preparation Course ===')
print(eda_df.groupby('test preparation course')[score_cols].mean().round(2))
```

```
▼ ... === Average Scores by Test Preparation Course ===
```

	math score	reading score	writing score
test preparation course			
completed	69.77	73.89	74.42
none	64.29	66.85	64.86

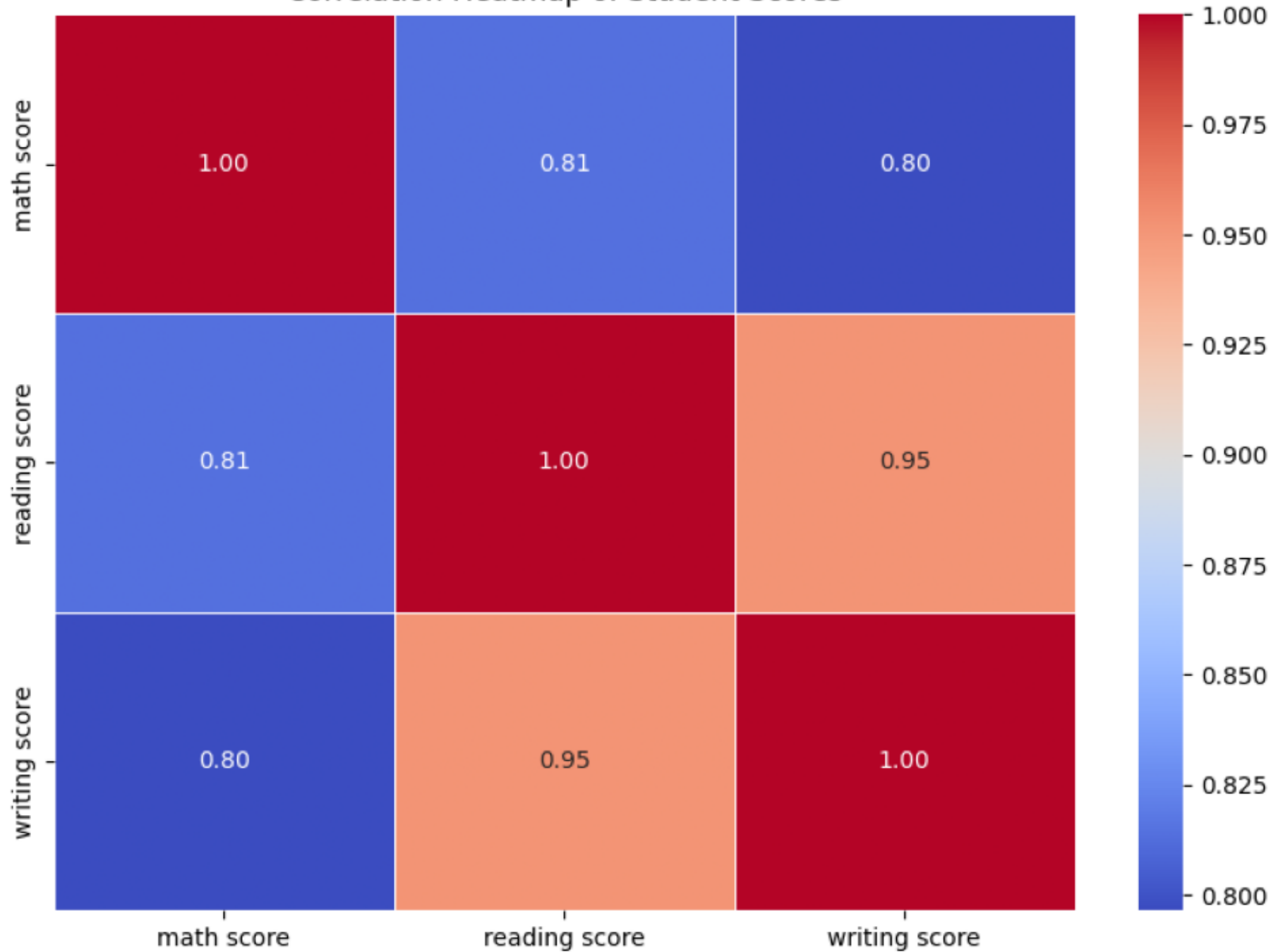
```
[26]  # Correlation Matrix
✓ 0s print('=== Correlation Matrix ===')
corr_matrix = eda_df[score_cols].corr()
print(corr_matrix.round(2))
```

```
▼ === Correlation Matrix ===
```

	math score	reading score	writing score
math score	1.00	0.81	0.80
reading score	0.81	1.00	0.95
writing score	0.80	0.95	1.00

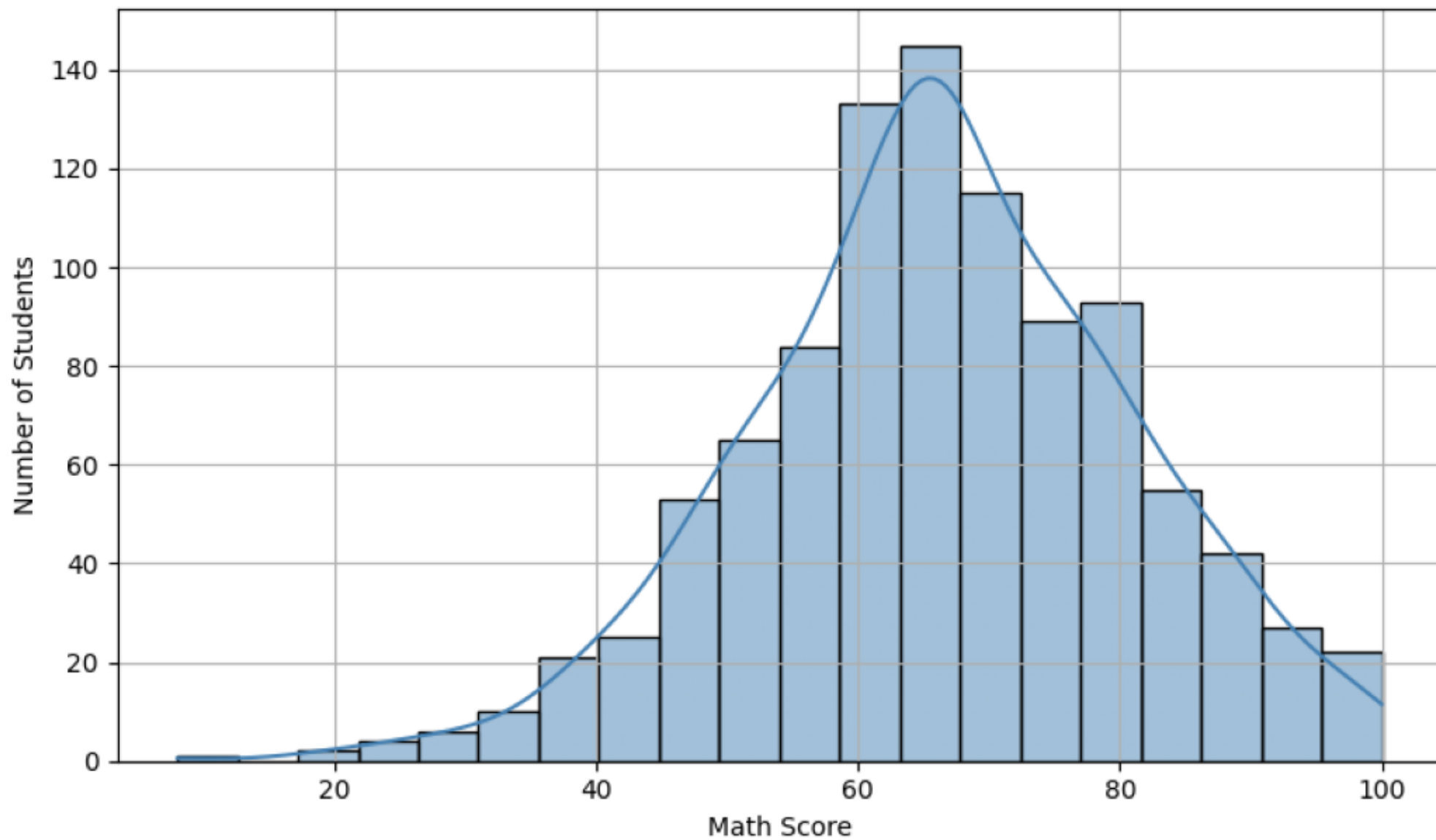
```
[28] # Seaborn Heatmap
```

Correlation Heatmap of Student Scores

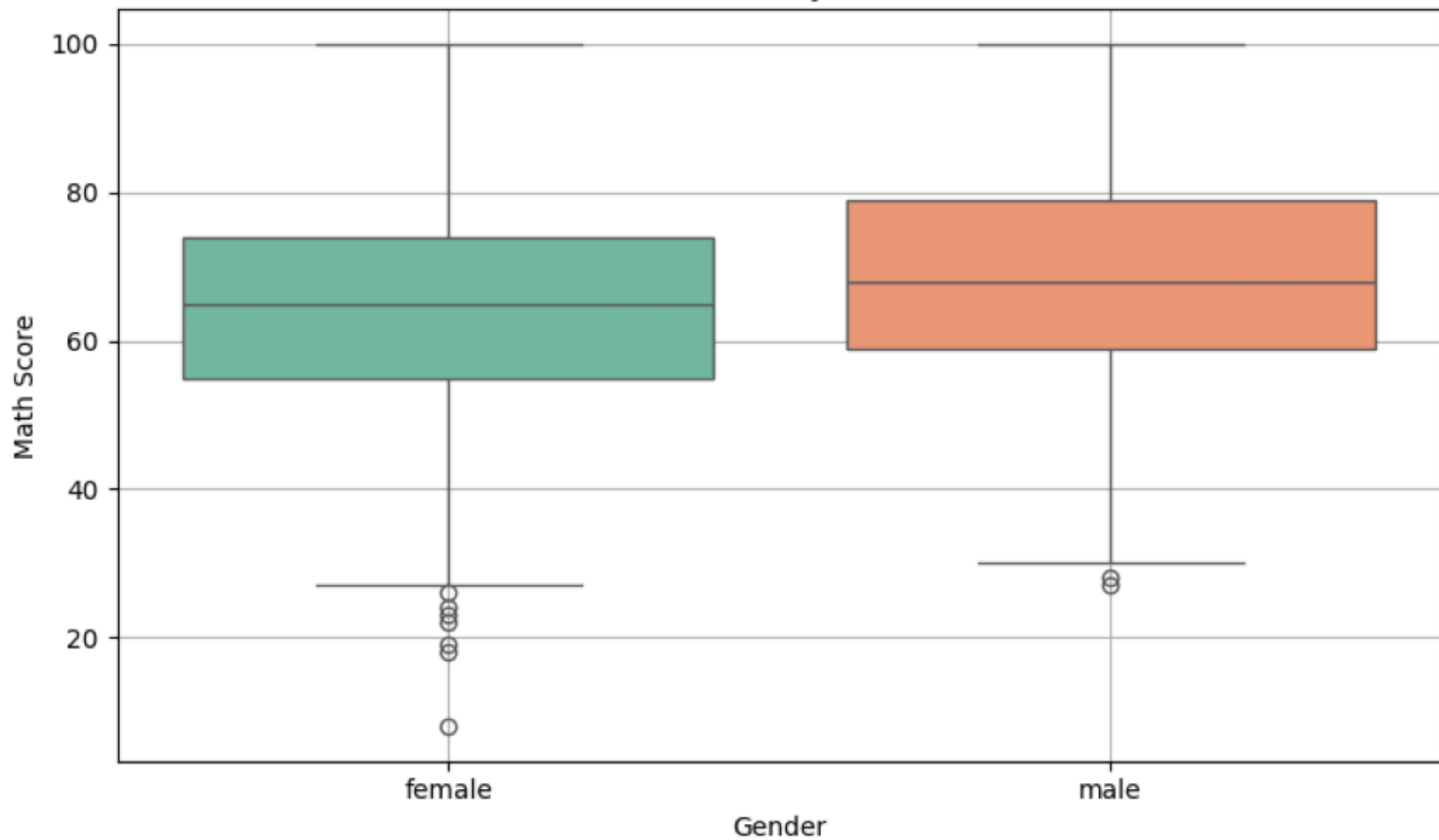


1. GENDER DIFFERENCES:
Female students consistently outperform male students in reading and writing scores. However, male students tend to score slightly higher in math. This pattern suggests a gender-based difference in subject preference or teaching approach. Schools could benefit from gender-targeted strategies to bridge these gaps.
2. PARENTAL EDUCATION IMPACT:
Students whose parents hold a master's or bachelor's degree score noticeably higher in all three subjects compared to students whose parents only completed high school. This highlights the strong influence of family educational background on student performance and suggests a cycle of educational advantage.
3. TEST PREPARATION COURSE:
Students who completed the test preparation course scored significantly higher across all subjects compared to those who did not complete it. This confirms that structured preparation directly and measurably improves academic performance. Encouraging more students to complete prep courses could raise overall scores.
4. RACE/ETHNICITY:
Group E students have the highest average scores across all subjects, while Group A students tend to score the lowest. This could reflect differences in access to educational resources, support systems, or socioeconomic factors.
5. CORRELATION BETWEEN SCORES:
Reading and writing scores are very strongly correlated (0.95+), meaning students who are good at reading are almost always good at writing too. Math shows a moderate correlation (~0.82) with both, suggesting these skills share some foundation but are more independent than reading and writing.
6. LUNCH TYPE (SOCIOECONOMIC INDICATOR):
Students with standard lunch score higher on average than students with free/reduced lunch. This socioeconomic indicator suggests that financial background plays a role in academic outcomes, possibly through access to tutoring and resources.

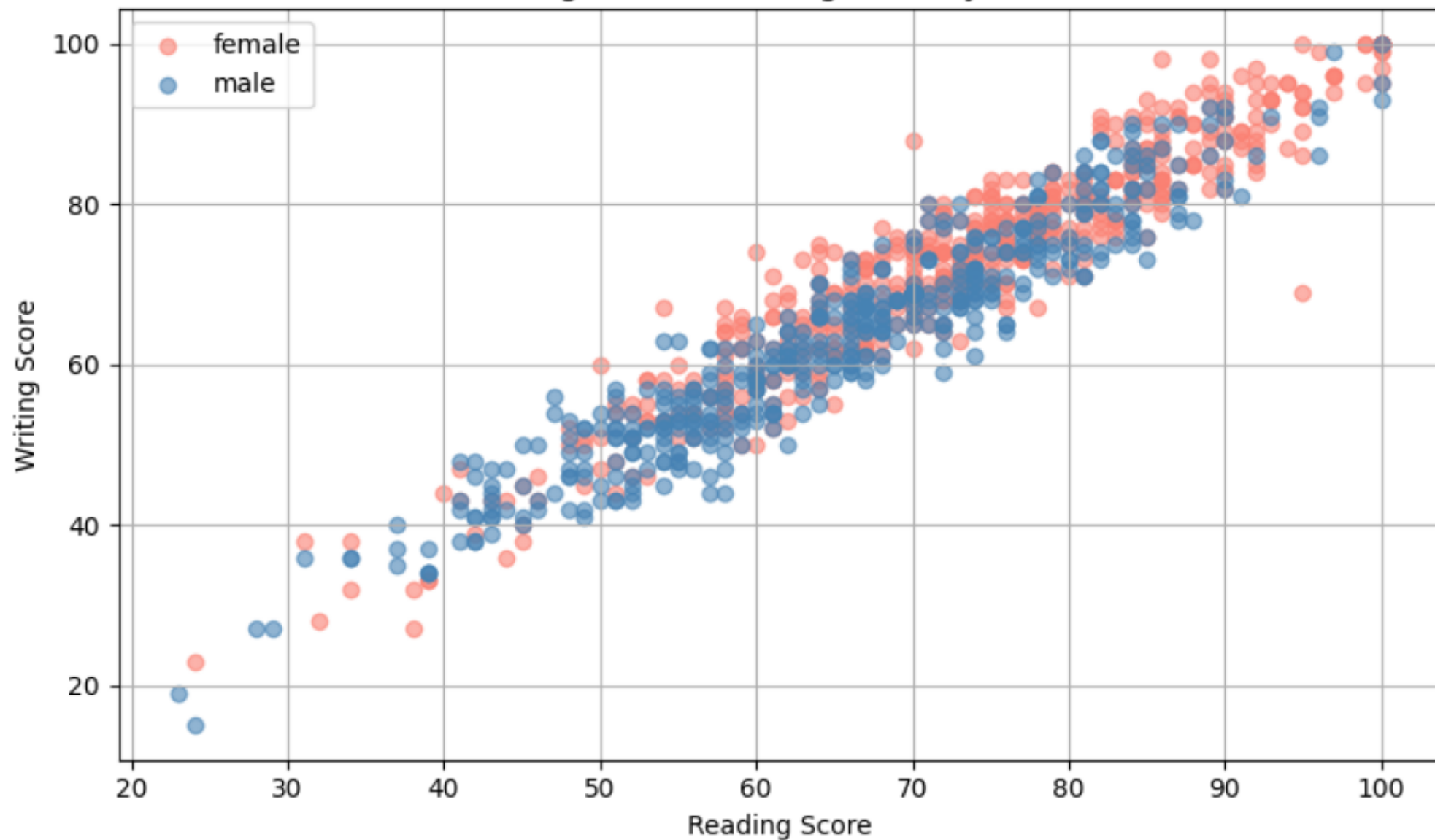
Distribution of Math Scores



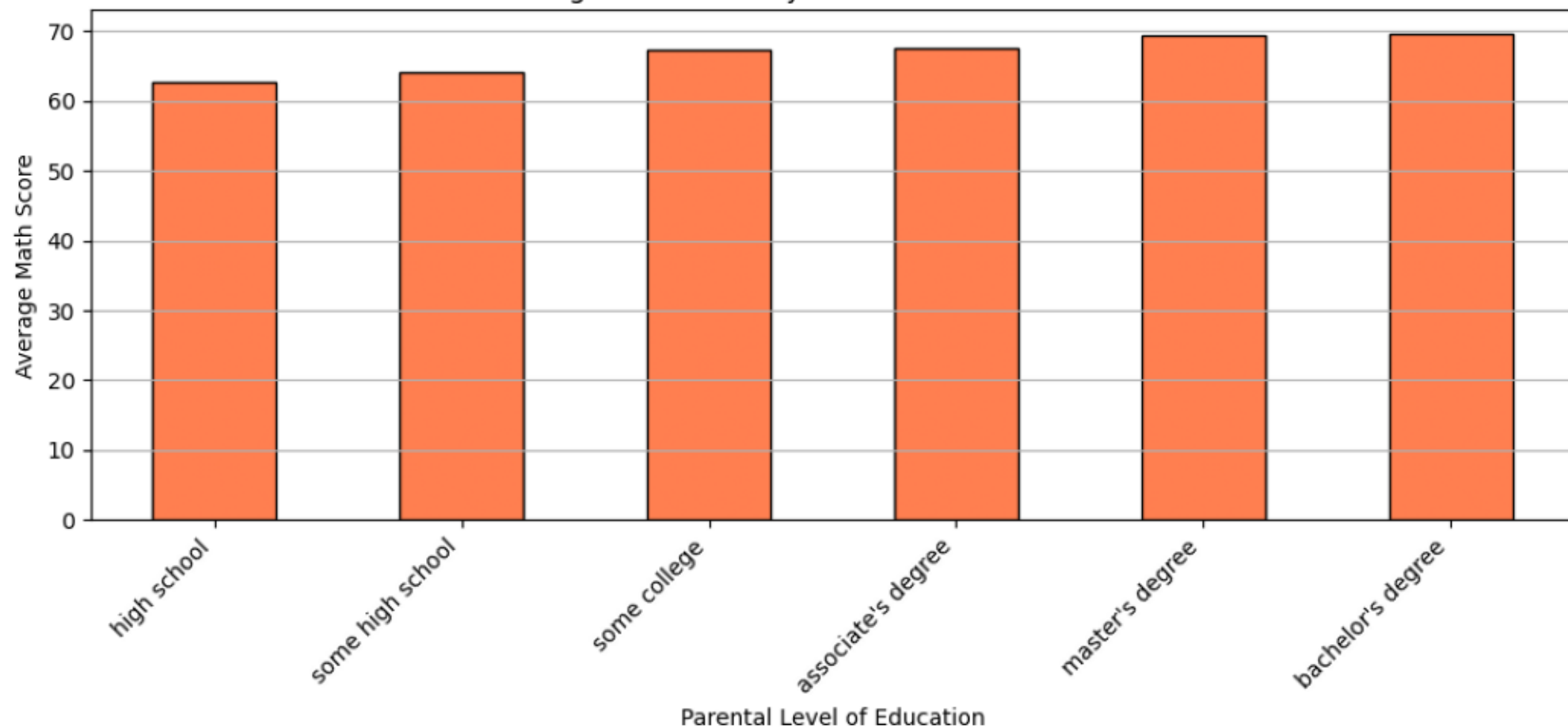
Math Score by Gender



Reading Score vs Writing Score by Gender



Average Math Score by Parental Level of Education



Student Performance - Combined Visualization

